One posterior, several decision problems

One supplied joint posterior
$\Pi_s(\theta \mid y)$

Estimate values

Target θ_i

Loss Squared value error

Output Posterior mean
$E(\theta_i \mid y)$

Estimate ranks

Target $R_i^{\downarrow}$

Loss Squared rank error

Output $E(R_i^{\downarrow} \mid y)$
May be noninteger;
No permutation required

Select a fixed top set

Target $T_b(\theta)$

Loss Top-set errors;
fixed size b

Output Top membership
probabilities; b units
No within-set order

Report common grades

Target Pairwise directions

Loss Direction errors
and withholding

Output $(B_1, \ldots, B_G)$
Coherent partition;
within-grade silence

Same scalar posterior; no latent ties.
Value loss: finite second moments.
Top-set size b is fixed.
Branches specify alternative tasks.